Options Mastery: Theory to Trading #1

Introduction to Black-Scholes Model

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Definition of European Option

- A European call (put) option is a contract that gives its owner the right, but not the obligation, to buy (sell) an underlying asset at a specified strike price by a specified expiry time.

- Three key attributes that characterize a European option are
  - Option type: Call or Put
  - Strike price
  - Expiry time

- The payoff of a European option with a strike price K and expiry time T is given by

$$\Psi(S_T) = \begin{cases} (S_T - K)_+ & \text{if Call} \\ (K - S_T)_+ & \text{if Put} \end{cases}$$

where S_T is the underlying asset's price at the expiry time.

Assumptions of Black-Scholes Model

1. There exists a money market account that grows at a constant risk-free interest rate, r.
2. The underlying asset price process, S_t, follows a geometric Brownian motion.
3. The underlying asset pays a constant dividend yield, q, continuously.
4. A market participant can buy or (short) sell any amount of an asset.
5. There are no transaction costs (frictionless market).
6. There is no arbitrage opportunity.

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Remark

In practice, these assumptions are often violated.

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Dynamics of Underlying's Price Process

Geometric Brownian Motion

$$\mathrm{d}S_t = \mu S_t \mathrm{d}t + \sigma S_t \mathrm{d}W_t^{\mathbb{P}}$$

- t is a time in years.
- $W_t^{\mathbb{P}}$ is a Brownian motion under $\mathbb{P}$ measure.
- $\mathscr{F}_t$ is the natural filtration (information) with respect to $W_t^{\mathbb{P}}$.
- μ is the annualized expected return: $\mu = \frac{1}{t} \log \left(\mathbb{E}^{\mathbb{P}} \left[\frac{S_t}{S_0} \middle| \mathscr{F}_0 \right] \right)$
- σ, usually called volatility, is the annualized standard deviation of log return: $\sigma^2 = \frac{1}{t} \mathrm{Var} \left(\log \left(\frac{S_t}{S_0} \right) \right)$

Properties of Underlying's Price Process

1. **Log Normality** $\log\left(\frac{S_t}{S_0}\right)\bigg|\mathscr{F}_0 \sim \mathscr{N}\left(\left(\mu-\frac{1}{2}\sigma^2\right)t, \sigma^2 t\right)$
2. **Independent Return** $\frac{S_{t_3}}{S_{t_2}} \perp\!\!\!\perp \frac{S_{t_1}}{S_{t_0}}$ if $t_0 < t_1 \leq t_2 < t_3$
3. **Markov** $S_t|\mathscr{F}_0 \overset{d}{=} S_t|S_0$
4. **Time Homogeneity** $\frac{S_{t+u}}{S_t}\bigg|\mathscr{F}_t \overset{d}{=} \frac{S_u}{S_0}\bigg|\mathscr{F}_0$
5. **Continuous Path** S_t is a continuous function of time.

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Remark

In practice, these properties are frequently observed to be invalid.

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Itô's Lemma

For a function $f \in C^{1,2}$,

$$\begin{aligned} \mathrm{d}f(t,S_t) &= \frac{\partial f}{\partial t}\mathrm{d}t + \frac{\partial f}{\partial S}\mathrm{d}S_t + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}(\mathrm{d}S_t)^2 \\ &= \left(\frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}S_t^2\sigma^2\right)\mathrm{d}t + \frac{\partial f}{\partial S}\mathrm{d}S_t \end{aligned}$$

Remark

Some useful multiplication rules $(\mathrm{d}t)^2 = \mathrm{d}t\mathrm{d}W_t = 0$ and $(\mathrm{d}W_t)^2 = 0$

$$\Rightarrow (\mathrm{d}S_t)^2 = S_t^2\sigma^2\mathrm{d}t$$

Assumptions & Motivation

Suppose

- There exists a unique price of an option, denoted by V.
- The price, $V = V(t, S_t)$, is a function of t and S_t.
- $V \in C^{1,2}$

Since the price depends on t and S_t, the underlying asset S_t is its only source of risk (randomness).

If we hedge the risk by holding some position of S_t, we can construct a riskless portfolio that should be identical to the money market account.

Riskless Option Portfolio

Let us consider a self-financing portfolio, Π:

- $\Pi_0 = 0$.
- Long the option.
- Short $\frac{\partial V}{\partial S_t}$ number of the underlying asset.
- The rest is invested in the money market account.

A portfolio is self-financing if there is no external infusion or withdrawal of money.

Portfolio Composition

	Cash Position	Cash Flow over $\mathrm{d}t$
Underlying Asset	$-\frac{\partial V}{\partial S} S_t$	$-\frac{\partial V}{\partial S} (\mathrm{d}S_t + qS_t \mathrm{d}t)$
Money Market	$\Pi_t - V_t + \frac{\partial V}{\partial S} S_t$	$r(\Pi_t - V_t + \frac{\partial V}{\partial S} S_t)\mathrm{d}t$
Option	V_t	$\mathrm{d}V_t$

Cash Flow of Option Portfolio

Using Itô's lemma for $\mathrm{d}V_t$, the cash flow of Π over $\mathrm{d}t$ is given by

$$\mathrm{d}\Pi_t = \left(r\Pi_t - rV_t + (r-q)\frac{\partial V}{\partial S} S_t + \frac{\partial V}{\partial t} + \frac{1}{2}\frac{\partial^2 V}{\partial S^2} S_t^2 \sigma^2 \right) \mathrm{d}t \tag{1}$$

Since this portfolio is riskless (no $\mathrm{d}W_t^{\mathbb{P}}$ term), it should grow at the same rate as the money market account. If not, it would create an arbitrage opportunity.

$$\mathrm{d}\Pi_t = r\Pi_t \mathrm{d}t \quad \Rightarrow \quad \Pi_t = e^{rt}\Pi_0$$

Since $\Pi_0 = 0$, $\Pi_t = 0$ and $\mathrm{d}\Pi_t = 0 \quad {}^\forall t \in [0, T]$.

Plugging in $\mathrm{d}\Pi_t = \Pi_t = 0$ into Eq. (1) yields the Black-Scholes PDE.

Black-Scholes PDE

Black-Scholes Partial Differential Equation

$$\begin{cases} (r-q)\frac{\partial V}{\partial S}S_t + \frac{\partial V}{\partial t} + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}S_t^2\sigma^2 = rV_t \\ V(T,S_T) = \Psi(S_T) \quad {}^\forall S_T \in (0,\infty) \end{cases}$$

- V doesn't depend on μ.
- The PDE holds for any European derivatives in the Black-Scholes world.
- The PDE can be rewritten as a form of the heat equation.

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Feynman-Kac Formula

The solution to the Black-Scholes PDE can be expressed as an expectation using the Feynman-Kac formula:

$$V_t = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}\left[\Psi(S_T)|S_t\right]$$

where

- $\mathrm{d}S_t = (r-q)S_t\mathrm{d}t + \sigma S_t \mathrm{d}W_t^{\mathbb{Q}}$.
- $W_t^{\mathbb{Q}}$ is a Brownian motion under $\mathbb{Q}$ measure.

$\mathbb{Q}$ measure is called a risk-neutral measure.

Remark

The price of any European derivative, if it exists, in the Black-Scholes world is equal to the discounted expectation of payoff under $\mathbb{Q}$.

Solution to Black-Scholes PDE

Black-Scholes Formula

$$V_t = \begin{cases} \Phi(d_1)e^{-q(T-t)}S_t - \Phi(d_2)e^{-r(T-t)}K & \text{if Call} \\ \Phi(-d_2)e^{-r(T-t)}K - \Phi(-d_1)e^{-q(T-t)}S_t & \text{if Put} \end{cases}$$

where

$$\begin{cases} d_1 = \frac{\log(S_t/K)+\left((r-q)+\frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}} \\ d_2 = d_1 - \sigma\sqrt{T-t} \end{cases}$$

and Φ is the cumulative distribution function of the standard normal distribution.

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From Black-Scholes to Black-76

- There are 5 parameters to price an option using the Black-Scholes model: S_t, K, T, σ, r, and q.
- In practice, estimating or predicting the dividend yield, q, may be challenging.
- One can get around this problem by using the Black-76 model.
- The Black-76 model takes the fair strike of a forward contract with the dividend yield q already priced in.

Black-76 Model

Black-76 Formula

$$V_t = \begin{cases} e^{-r(T-t)}\left[\Phi(d_1)F_t - \Phi(d_2)K\right] & \text{if Call} \\ e^{-r(T-t)}\left[\Phi(-d_2)K - \Phi(-d_1)F_t\right] & \text{if Put} \end{cases}$$

where

$$\begin{cases} d_1 = \frac{\log(F_t/K)+\frac{1}{2}\sigma^2(T-t)}{\sigma\sqrt{T-t}} \\ d_2 = d_1 - \sigma\sqrt{T-t} \end{cases}$$

and F_t is the fair strike of a forward contract that matures at T.

In Black-Scholes world, $F_t = S_t e^{(r-q)(T-t)}$. Using this, one can easily show that the Black-Scholes and Black-76 formulae are equivalent.

Reference

- Björk, T. (2009). Arbitrage theory in continuous time. Oxford University Press.
- Shreve, S. E. (2004). Stochastic calculus for finance II: Continuous-time models. Springer.

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